Particle in a box with a single barrier

IMMA MONCUSÍ LASO

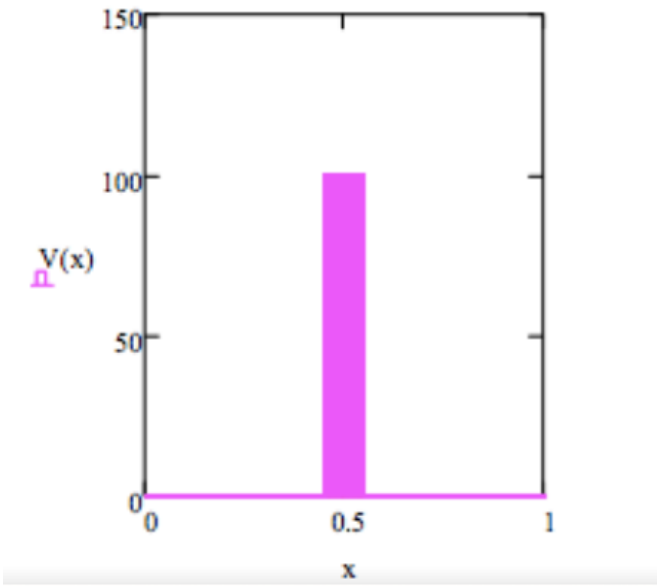
Initial conditions

Reference:

V=100

L=1

Lbarrier= 0.1

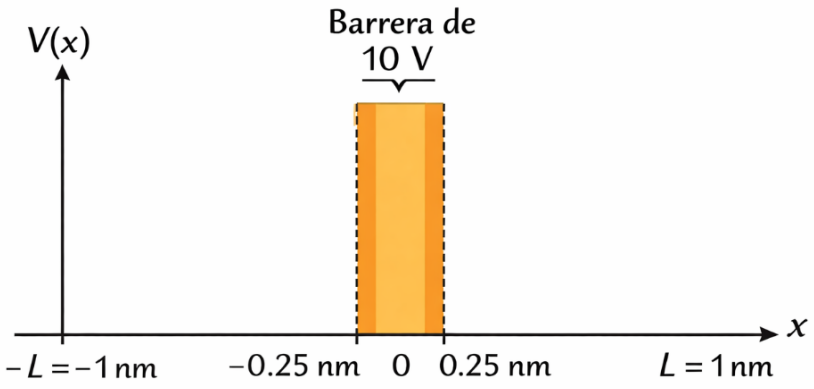


P1:

V=10

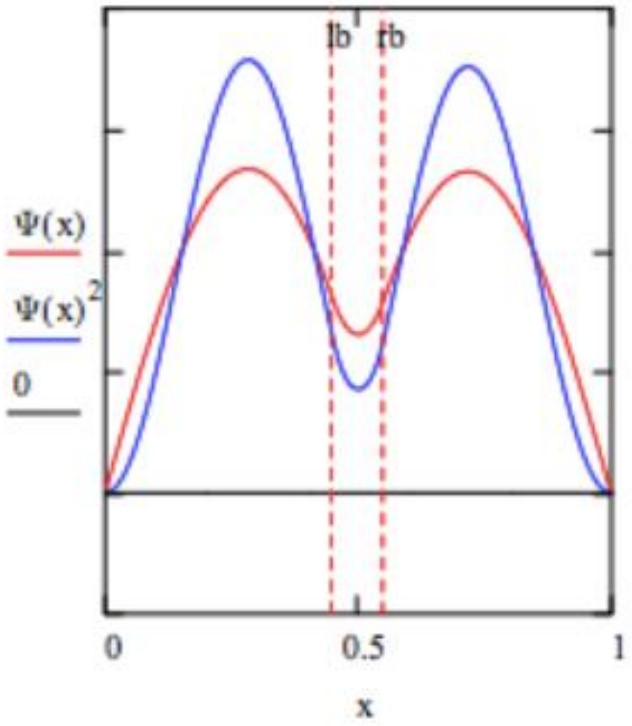
L=2

Lbarrier= 0.5

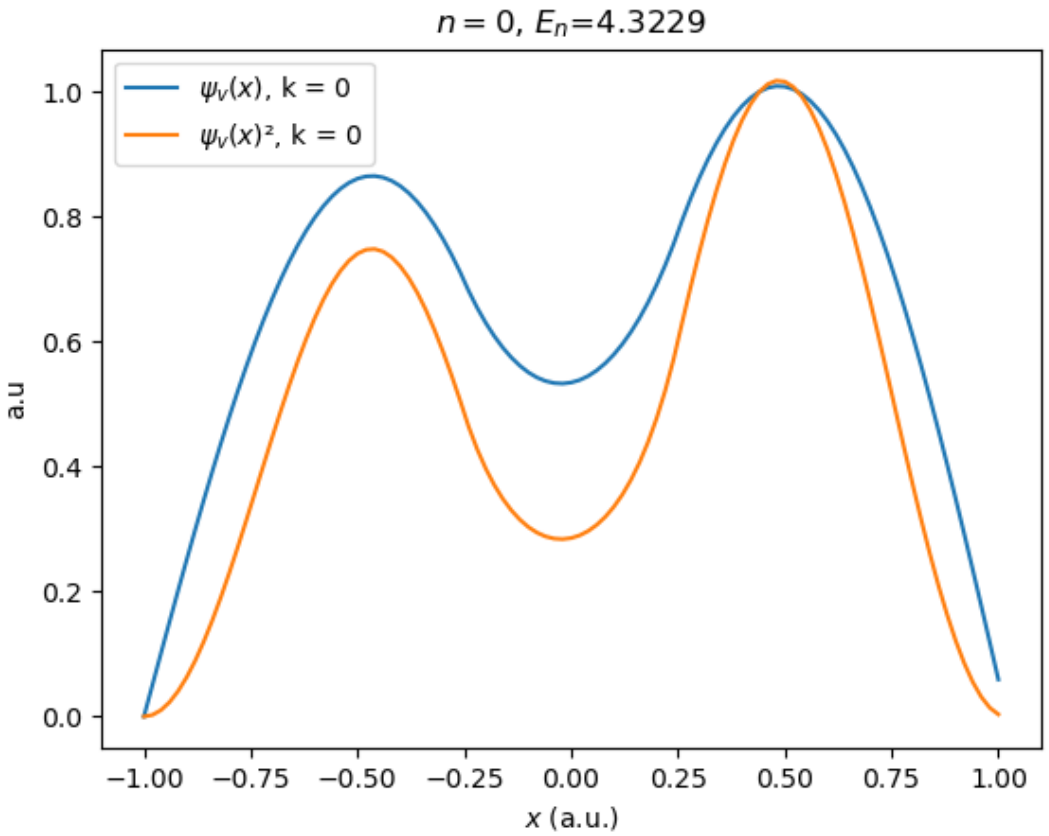


Caixa 1D: $-L$ a L, $Lt = 2$ nm

Reference vs result

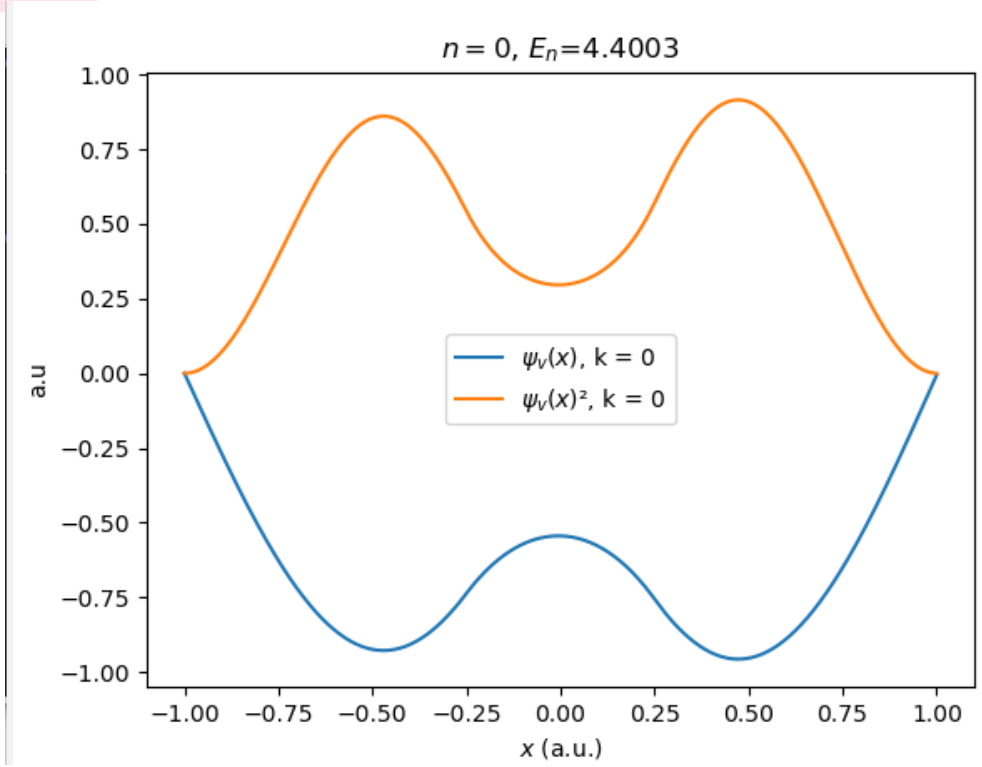


Reference



h=0.02

Reference vs result



h=0.002

Something happened, maybe this?

```
def getV(x):

    if (abs(x)<=0.25):
        potvalue = 10.0
    else:
        potvalue =0.0
    return potvalue
```

The probability makes sense, but psi is like it is multiplied by "-1"

Solution intentSolution intent

```
#Calculation of normalised Wave Functions
for k in range(0,len(w)):
    psi[k] = vs[:,k]
    #psi2[k] = vs[:,k]
    integral = h*np.dot(psi[k],psi[k])
    psi[k] = psi[k]/integral**0.5 #aquesta és la fórmula per
    normalitzar la funció diona
    psi2[k]=psi[k]**2
    if psi[k]<0:
        psi[k]*=-1
    return psi[k]
```

```
SyntaxError: 'return' outside function
(base) 1667196@uab.cat@CCBC-pc1d-12-L:~/Documents$ python p1a.py
  File "/home/UAB/1667196/Documents/p1a.py", line 91
    return psi[k]
    ^^^^^^^^^^^^^
```